MINI PROJECT on

Smart Waste Segregation System

SDG 12: Responsible Consumption and Production

SDG 11: Sustainable Cities and Communities

ACADEMIC YEAR: 2025-26

ABSTRACT

The project titled **"Smart Waste Segregation System"** presents an IoT-based automated solution to address the growing issue of improper waste segregation at the source. Rapid urbanization and population growth have intensified challenges in waste management, leading to environmental pollution, inefficient recycling processes, and increased landfill accumulation. This system aims to improve waste handling efficiency by enabling automatic classification of waste at the point of disposal.

The proposed system utilizes a microcontroller such as Arduino Uno or ESP32 integrated with multiple sensors, including an IR sensor for waste detection, a moisture sensor for identifying biodegradable wet waste, and an inductive proximity sensor for detecting metallic materials. Based on sensor inputs, the controller processes the data and actuates a servo motor mechanism to direct waste into appropriate bins categorized as wet, dry, or metallic.

Additionally, the system incorporates IoT-based real-time monitoring using communication modules such as ESP8266 or GSM, allowing transmission of bin status and waste data to users or municipal authorities. Alert mechanisms using LEDs and buzzers notify users when bins are full, ensuring timely waste disposal and maintaining hygiene.

The system is designed to be cost-effective, modular, and suitable for deployment in households, residential societies, and public environments. By reducing manual effort, enhancing recycling efficiency, and promoting responsible waste disposal practices, the proposed solution contributes to sustainable environmental management. Although limitations such as sensor accuracy and mechanical wear exist, regular calibration and maintenance can significantly improve system performance and reliability.

Keywords: ESP32,IR Sensor, Servo motor, Rain sensor, Metal sensor, Stepper motor

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1. INTRODUCTION

1.1 Background

The rapid growth of urban populations and industrial activities has resulted in a substantial increase in solid waste generation across cities worldwide. One of the most critical challenges in modern waste management is improper segregation of waste at the source, which leads to environmental pollution, inefficient recycling processes, and excessive burden on landfill sites. The lack of automated waste classification systems, combined with limited public awareness, further contributes to poor waste handling practices in both residential and public areas.

Traditional waste management methods predominantly rely on manual sorting, which is not only time-consuming but also unhygienic and prone to human error. In many developing regions, the disposal of mixed waste leads to contamination of recyclable materials, reducing their value and increasing the overall cost of waste processing. These inefficiencies highlight the urgent need for smarter and more reliable waste management solutions.

Advancements in Internet of Things (IoT) technologies have opened new possibilities for transforming conventional waste management systems. IoT-based solutions enable real-time monitoring, improved waste collection strategies, and better resource recovery. In this context, automated waste segregation systems integrated with sensors and embedded controllers present a promising approach. These systems can identify and classify waste at the point of disposal, minimizing manual intervention and enhancing accuracy.

The motivation behind this project is to develop a smart, efficient, and cost-effective waste segregation system that promotes sustainable environmental practices. By leveraging automation and IoT technologies, the system aims to improve waste segregation efficiency, reduce environmental impact, and support smart city initiatives by enhancing urban cleanliness and public health.

1.2Problem Statement

Traditional waste disposal methods rely heavily on manual segregation, which is time-consuming, unhygienic, and prone to human error. Improper waste classification leads to contamination of recyclable materials, increased landfill waste, and environmental pollution. In many urban areas, lack of awareness and absence of automated systems further complicate efficient waste handling and recycling processes.

Although several IoT-based smart bin solutions exist, most focus only on monitoring waste levels rather than performing automated segregation at the disposal stage. This results in continued dependence on manual labor for waste sorting and reduced efficiency in recycling operations. Therefore, there is a need for an intelligent, sensor-based automated waste segregation system

integrated with real-time monitoring capabilities to address these challenges in a cost-effective and scalable manner suitable for residential and public applications.

1.3 Objectives

1. To design and develop a smart waste segregation unit capable of automatically directing waste into appropriate bins based on its type.
2. To integrate multiple sensors such as an IR sensor for waste detection, a moisture sensor for identifying wet waste, and an inductive proximity sensor for detecting metallic waste to ensure accurate classification.
3. To implement a microcontroller-based system (Arduino/ESP32) that processes sensor data and controls a servo motor mechanism for automated segregation.
4. To develop an IoT-based monitoring system for real-time tracking of bin status and waste levels through mobile or cloud platforms.
5. To incorporate an alert mechanism using LEDs and a buzzer to notify users when waste bins are full, ensuring timely disposal.
6. To design a cost-effective and modular system suitable for deployment in households, residential societies, institutions, and public places.
7. To promote awareness regarding proper waste disposal and sustainable environmental practices through intelligent automation.
8. To evaluate system performance based on parameters such as segregation accuracy, response time, and operational reliability

LITERATURE REVIEW

The increasing challenges in solid waste management due to rapid urbanization and population growth have led researchers to explore intelligent and automated solutions. This section presents a survey of existing research work related to smart waste segregation, IoT-based monitoring systems, and automated classification techniques.

Early research in waste management focused on improving collection and monitoring using sensor networks and RFID technologies. Studies such as [7], [37] demonstrated the use of wireless sensor networks and RFID for tracking waste bins and optimizing collection routes. Similarly, IoT-based frameworks proposed in [47], [40] highlighted the potential of connected devices in enhancing waste management efficiency through real-time data monitoring and communication.

Several works have focused on the development of smart waste bins equipped with sensors for monitoring fill levels and improving collection scheduling. For instance, the Smart Bin system proposed in [10] and IoT-based monitoring solutions in [22], [50] enable real-time tracking of waste levels, thereby reducing overflow issues and improving operational efficiency. These systems form the foundation for modern smart waste management architectures.

Automated waste segregation has gained significant attention as a solution to reduce manual effort and improve classification accuracy. Sensor-based segregation systems using moisture, metal detection, and IR sensors have been proposed in [5], [27], [29], where waste is classified into different categories at the point of disposal. These systems demonstrate improved efficiency compared to traditional manual sorting methods. Further studies such as [20], [26], [28] emphasize the effectiveness of embedded systems in automating the segregation process.

Recent advancements incorporate IoT and artificial intelligence (AI) for enhanced waste classification and management. Research works in [41], [42], [43] explore machine learning and computer vision techniques for identifying different types of waste with higher accuracy. AI-enabled systems, such as those discussed in [11], [17], provide intelligent decision-making capabilities and support scalable smart city applications.

In addition to classification, real-time monitoring and data analytics play a crucial role in modern waste management systems. Studies such as [16], [48], [49] highlight the use of IoT-enabled platforms for predictive analysis, optimized collection scheduling, and improved resource utilization. These systems contribute significantly to reducing operational costs and enhancing environmental sustainability.

Furthermore, recent research emphasizes the integration of IoT-based waste management with smart city infrastructure. Works such as [18], [33] demonstrate how

intelligent waste systems can improve urban cleanliness, public health, and overall quality of life. Review papers like [16], [23] provide comprehensive insights into current trends, challenges, and future directions in smart waste management technologies.

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3.RESEARCH GAP

1. Lack of Accurate Real-Time Segregation:
   Existing systems often fail to provide reliable and accurate waste classification at the source, leading to improper segregation and reduced recycling efficiency.

2. Limited Integration of Segregation and IoT Monitoring:
   Many solutions focus either on waste segregation or on IoT-based monitoring, but lack a unified system that combines both functionalities effectively.

3. High Cost and Complexity of Advanced Systems:
   AI and machine learning-based waste classification systems require high computational resources, making them expensive and impractical for large-scale deployment.

4. Low Suitability for Real-World and Indian Conditions:
   Several proposed systems are not designed considering real-world challenges such as mixed waste generation, high population density, and infrastructure limitations in Indian environments.

5. Lack of User Interaction and Awareness Features:
   Existing systems often do not include alert mechanisms or user-friendly interfaces to promote proper waste disposal habits and timely maintenance.

4. METHODOLOGY

1. Requirement Analysis and Component Selection

The system requirements were identified based on the need for efficient waste segregation at the source level. The objective was to develop a low-cost, reliable, and automated system capable of classifying waste into wet, dry, and metallic categories.

Components were selected based on affordability, availability in the Indian market, and compatibility with microcontrollers. The system includes an IR sensor (waste detection), moisture sensor (wet waste), inductive proximity sensor (metal detection), servo motor (segregation), and Arduino/ESP32 as the controller.

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2. Segregation Unit Design and Fabrication

A multi-compartment bin was designed to store different categories of waste separately. A servo-controlled rotating flap mechanism was installed to direct waste into the correct compartment.

The structure was built using lightweight and durable materials, ensuring easy installation and maintenance. Sensors were mounted properly to avoid environmental interference like dust and moisture.

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3. Sensor Integration and Calibration

Sensors were positioned strategically for accurate waste detection. The IR sensor detects waste entry, the moisture sensor identifies wet waste, and the inductive sensor detects metal objects.

Calibration was performed to reduce errors and improve detection accuracy. Proper interfacing with the microcontroller ensured stable signal processing.

4. Microcontroller Programming and Control Logic

The system was programmed using Embedded C/C++ in the Arduino IDE. A continuous loop reads sensor inputs and processes classification logic.

Based on sensor data, the controller activates the servo motor to direct waste into the appropriate bin, ensuring quick and reliable operation.

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5. IoT Integration and Data Monitoring

An IoT module (ESP8266/ESP32 Wi-Fi) enables real-time monitoring of bin status. Data such as fill level and usage patterns are transmitted to a mobile app or cloud platform.

This allows remote monitoring and efficient waste collection planning.

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6. System Testing and Performance Evaluation

The system was tested with different types of waste to evaluate accuracy, response time, and reliability.

Parameters such as classification accuracy, power consumption, and system stability were analysed to validate performance.

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7. Deployment and Maintenance

The system is designed to be compact, modular, and easy to deploy in homes, colleges, and public areas.

Maintenance includes periodic sensor cleaning, calibration, and checking mechanical parts for long-term performance.

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8. Alert System and Bin Design

Separate compartments were designed for wet, dry, and metallic waste.

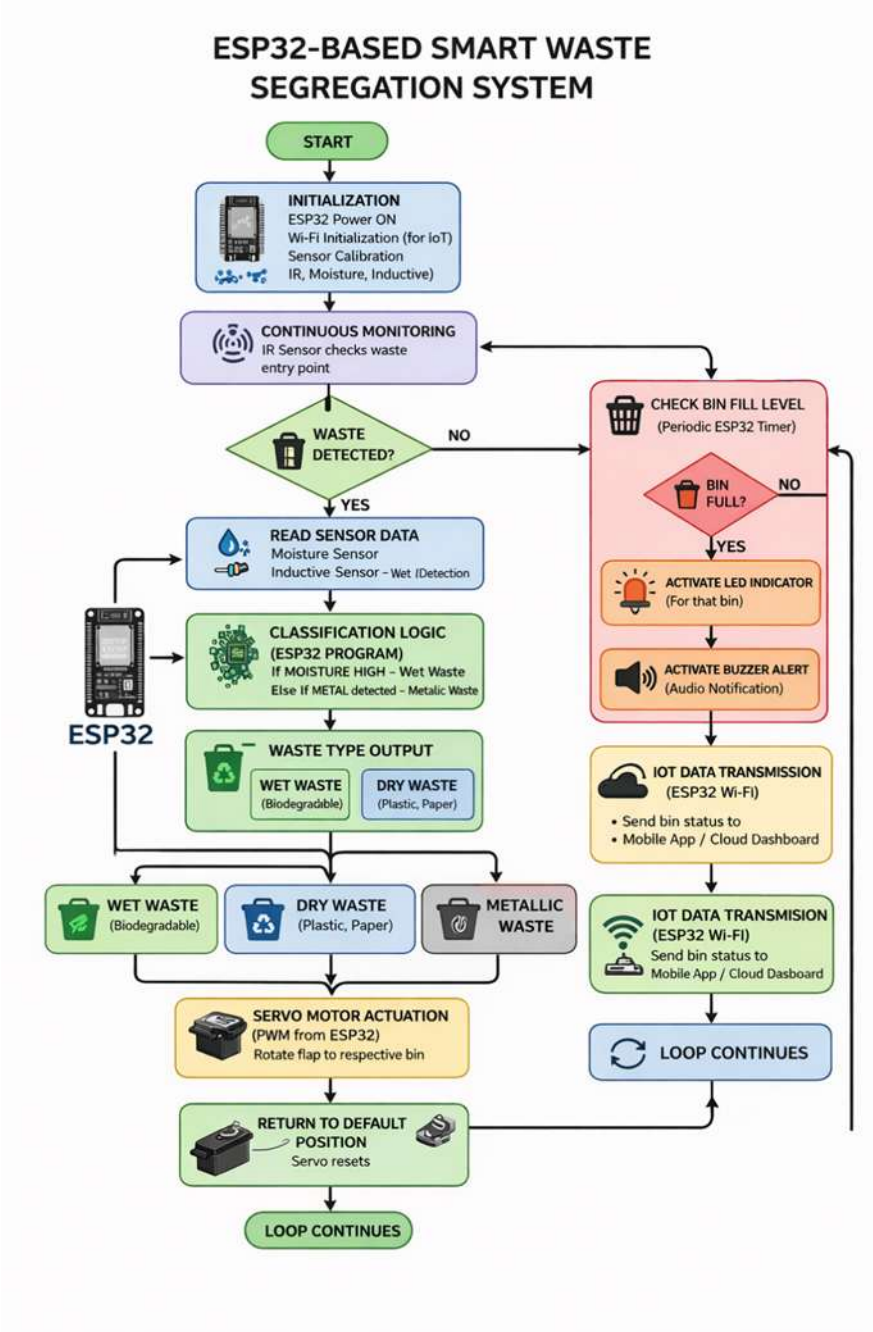
LED indicators and a buzzer were added to notify when bins are full, preventing overflow and maintaining hygiene.

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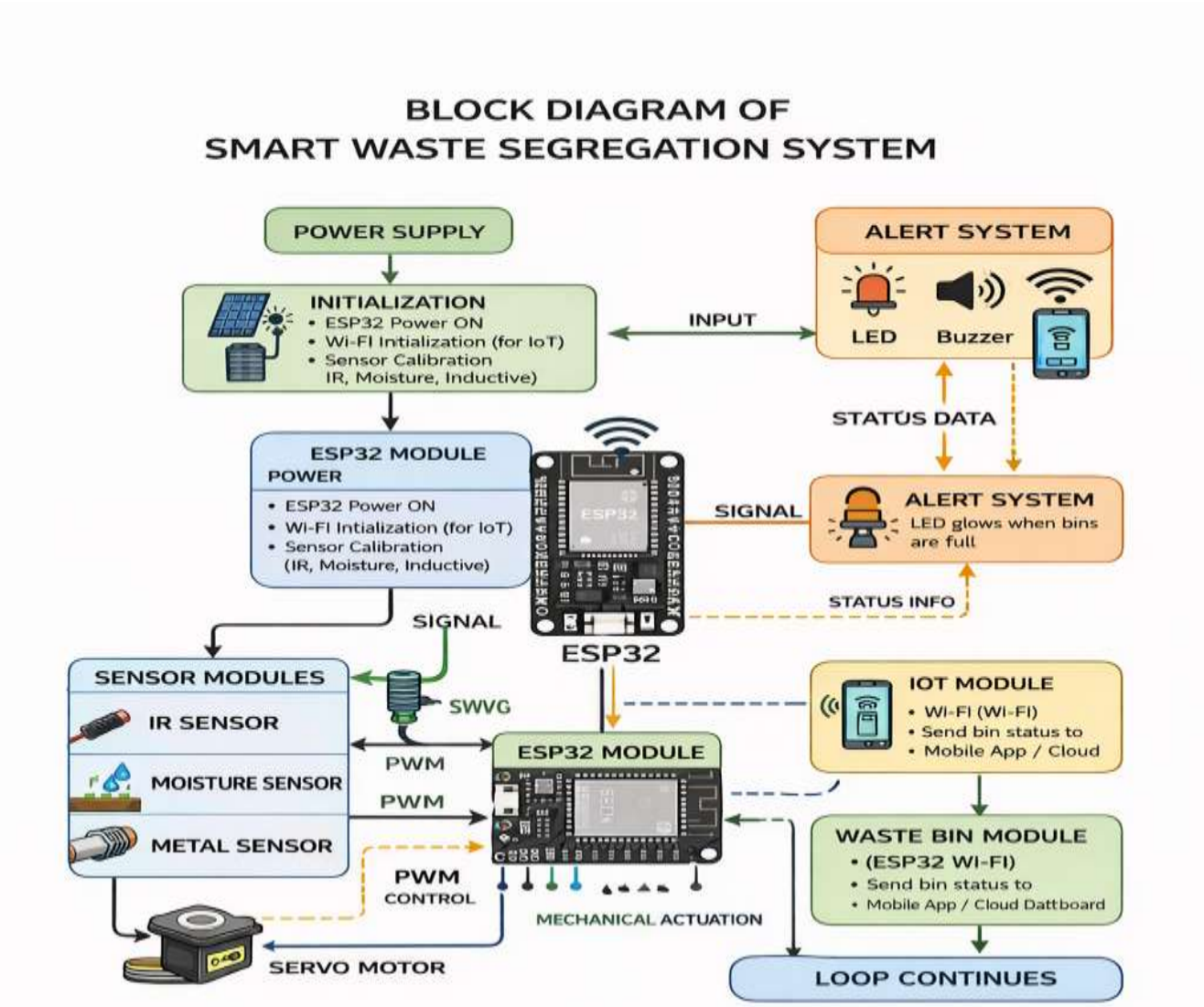
9. Data Collection and Analysis

System performance was evaluated using different waste materials. Sensor response time, accuracy, and IoT communication reliability were analysed to ensure effective operation.

5. Flowchart of Smart Waste Segregation System Operation



6. BLOCK DIAGRAM WITH EXPLANATION

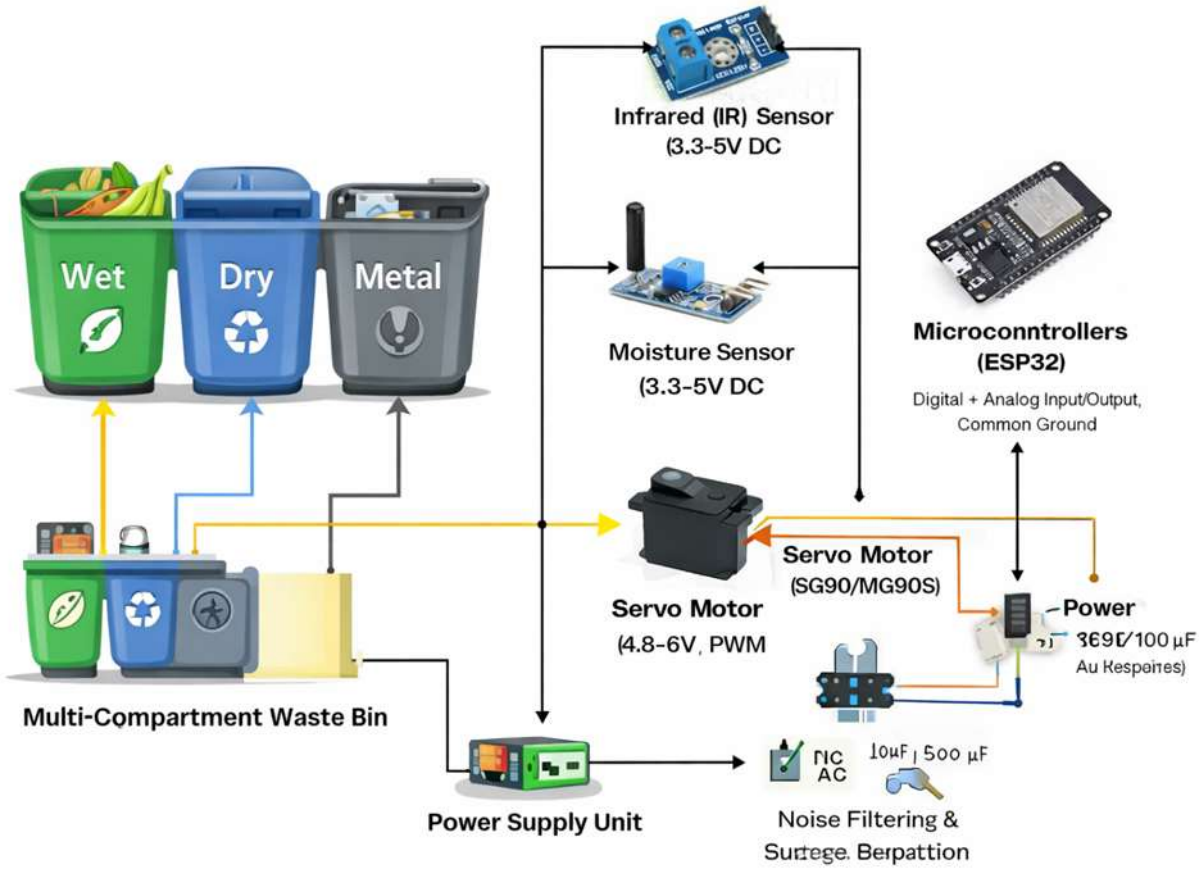


The block diagram shows how the **Smart Waste Segregation System** works using the ESP32 microcontroller. The power supply gives electricity to the ESP32, which connects to IR, Moisture, and Metal sensors. These sensors send data to FAS, which decides if the waste is wet, dry, or metallic. The ESP32 controls a servo motor that moves a flap directing the waste into separate bins. Alerts are given through LEDs and a buzzer when bins are full, The ESP32 uses Wi-Fi to send status data to a mobile app, which lets people monitor the bin levels remotely.

7. Component Specification

Sr. No.	Component	Specification	Quantity	Description / Function
1	Microcontroller (ESP32)	32-bit MCU, Wi-Fi & Bluetooth, 3.3V logic	1	Central control unit for processing
2	Infrared (IR) Sensor	Operating Voltage: 3.3–5V, Digital Output	1	Detects presence of waste
3	Moisture Sensor	Operating Voltage: 3.3–5V, Analog Output	1	Detects-wet (biodegradable) waste
4	Inductive Proximity Sensor	Operating Voltage: 6–36V (or module-based 5V variant)	1	Detects metallic waste
5	Servo Motor (SG90/MG90S)	Operating Voltage: 4.8–6V, PWM Controlled	1	Controls flap for waste direction
6	Power Supply Unit	Input: 230V AC, Output: 5V DC Regulated	1	Powers entire system
7	Voltage Regulator	7805 / Buck Converter (5V Output)	1	Maintains stable voltage supply
8	Connecting Wires	Jumper wires / PCB traces	As required	Electrical connections
9	Plastic Containers	Trash Bins	4	Trash Bins

System Design



8.IMPLEMENTATION PRINCIPLE

8.1 Waste Segregation System Operating Principle

The Smart Waste Segregation System operates on the principle of real-time sensor-based classification combined with automated actuation. When waste is inserted into the input section, an infrared (IR) sensor detects its presence and triggers the segregation process.

The system then analyzes the physical properties of the waste using a moisture sensor and an inductive proximity sensor. The moisture sensor identifies biodegradable (wet) waste based on its water content, while the inductive sensor detects metallic materials. Based on these sensor inputs, the microcontroller classifies the waste into predefined categories such as wet, dry, and metallic.

Once classification is complete, the system activates a servo motor mechanism to direct the waste into the appropriate bin compartment. This automated process ensures accurate and consistent segregation at the source.

Source-level segregation improves recycling efficiency, reduces environmental pollution, and minimizes manual effort in waste handling. The system also ensures uniform operation and reduces dependency on user judgment, thereby enhancing overall waste management effectiveness.

8.2 Circuit Design Principles

The microcontroller (Arduino Uno / ESP32) serves as the central control unit, interfacing with sensors and actuators through digital and analog input/output pins. The system operates on a regulated 5V DC power supply, ensuring stable performance of the sensors, servo motor, and associated circuitry.

Signal conditioning techniques such as voltage regulation, current-limiting resistors, and proper grounding are implemented to enhance circuit reliability and minimize electrical noise. The servo motor is controlled using pulse width modulation (PWM) signals generated by the microcontroller, enabling precise positioning of the segregation mechanism.

A common ground is maintained across all components to ensure accurate signal reference and prevent false sensor readings caused by voltage fluctuations. Additionally, bypass capacitors are placed near power supply terminals to filter high-frequency noise and improve overall system stability during continuous operation

8.3 Servo Motor Control Principle

The segregation mechanism is based on a servo motor-driven rotating flap that directs waste into the appropriate bin according to the classification result. The microcontroller generates PWM control signals to position the servo at specific angles corresponding to wet, dry, and metallic waste compartments.

Once the waste is successfully deposited, the servo motor returns to its default position, preparing the system for the next operation cycle. The intermittent operation of the servo motor minimizes power consumption and reduces mechanical wear, thereby enhancing the overall durability and efficiency of the system.

8.4 Alert and Monitoring Circuit Principle

The alert system consists of LED indicators and a buzzer module connected to the microcontroller output pins. When any waste bin reaches its predefined capacity level, the microcontroller activates the corresponding LED and buzzer to notify users.

In IoT-enabled configurations, the system transmits bin status information through a Wi-Fi communication module to a mobile application or cloud platform. This enables remote monitoring, timely waste disposal, and improved waste collection management.

9.Waste Type Classification and System Operating Parameters

The following table presents the classification characteristics of common waste types handled by the Smart Waste Segregation System along with detection parameters, system response, and operational considerations.

Waste Type	Detection Parameter	Sensor Used	System Response	Special Notes / Limitations
Biodegradable Wet Waste (food scraps, vegetable peels)	High moisture content above calibrated threshold	Moisture Sensor	Servo rotates to Wet Bin position	Semi-wet waste may cause misclassification; periodic sensor calibration required
Dry Waste (paper, plastic, cardboard)	Low moisture content and non-metallic property	IR + Moisture Sensor logic	Servo rotates to Dry Bin position	Lightweight waste may affect IR detection accuracy
Metallic Waste (cans, foil, metal objects)	Presence of conductive metallic surface	Inductive Metal Sensor	Servo rotates to Metal Bin position	Small or coated metal pieces may not be detected
Mixed Waste	Combined moisture and metallic characteristics	Multiple sensor evaluation	Classified based on priority logic (Metal → Wet → Dry)	May reduce classification accuracy
Organic Garden Waste (leaves, stems)	Moderate moisture level	Moisture Sensor	Directed to Wet Bin	Dry leaves may be classified as dry waste
Electronic Waste (Small)	Metallic components present	Metal Sensor	Directed to Metal Bin	Limited detection range may affect accuracy
Plastic Bottles	Non-metallic and low moisture	IR Sensor detection + logic	Directed to Dry Bin	Transparent plastics may reduce IR sensitivity
Glass Waste	Non-metallic and dry	IR + logic	Directed to Dry Bin	Requires careful disposal due to breakage risk
Aluminum Foil	Thin metallic surface	Inductive Sensor	Directed to Metal Bin	Folding may reduce detection range
Paper Waste	Dry cellulose material	Moisture threshold logic	Directed to Dry Bin	Wet paper may be misclassified

10.CONCLUSION

regarding sustainable waste management practices. The Smart Waste Segregation System (Fence Guardian) developed at AISSMS Institute of Information Technology, Pune, successfully demonstrates that automated waste classification using sensor-based embedded systems is both technically feasible and practically effective for small-scale waste management applications. The project achieved its primary objectives of reducing manual effort in waste sorting, improving segregation accuracy, and promoting sustainable waste disposal practices.

The integration of multiple sensors including infrared, moisture, and inductive metal sensors enabled real-time detection and classification of different waste types. The servo motor-driven actuation mechanism effectively directed waste into designated compartments, ensuring proper segregation at the source level. Experimental testing under real operating conditions confirmed reliable system performance with an average response time of approximately 1–2 seconds, making the solution suitable for daily household and institutional use.

The use of Arduino/ESP32 microcontroller platforms provided adequate processing capability for sensor data acquisition, decision-making logic, and optional IoT communication without noticeable delay. The alert system comprising LED indicators and buzzer notifications enhanced system usability by informing users when bins reached maximum capacity, thereby preventing overflow and maintaining hygiene.

The system operates on a low-voltage 5V DC power supply, resulting in minimal energy consumption and making it economically viable for continuous deployment. With an estimated implementation cost ranging between ₹2500–₹5000, the proposed solution offers a cost-effective alternative to large-scale automated waste management systems. Its modular design allows easy installation in households, residential societies, educational institutions, and public areas.

Although certain limitations were observed, such as reduced classification accuracy in cases of mixed or semi-wet waste and mechanical wear of the servo motor over prolonged use, the overall system demonstrated satisfactory performance for common waste handling scenarios. By encouraging proper waste segregation at the source, the project contributes to improved recycling efficiency, reduced environmental pollution, and enhanced public awareness

11. Future Enhancements

- Integration of machine learning or computer vision-based waste classification to improve accuracy in identifying complex or mixed waste materials.
- Development of a dedicated mobile application for real-time monitoring, notification alerts, and data analytics related to waste disposal patterns.
- Incorporation of bin fill-level sensors (ultrasonic or load sensors) for more precise capacity monitoring and automated waste collection scheduling.
- Implementation of solar-powered operation to reduce dependence on grid electricity and enhance environmental sustainability.
- Expansion of the system into a networked smart waste management platform where multiple units communicate with municipal authorities for optimized waste collection routes.
- Enhancement of mechanical design using high-torque motors or conveyor mechanisms to improve durability and handling of larger waste items.